

CardioScope — Interactive 3D Cardiovascular Risk Visualisation

Track A · Multimodal AI Hackathon 2026 · Team of 4 (M1–M4) Source: <https://github.com/vaibhavitej-a11y/CardioScope>

For decision support / educational purposes only — not a substitute for formal diagnostic imaging.

1. Overview

CardioScope predicts **overall CAD status** and **predicted stenosis probability** for the three major coronary arteries (LAD, LCX, RCA) from 54 routine clinical features, then maps those predictions onto an interactive 3D heart: each vessel is colour-ramped by its predicted risk, selectable for a detail view, and explained with SHAP contributions. The system answers two questions a table of numbers cannot: *how sick is this patient* and *where is the risk* — in one screen.

Architecture. Three layers, one repo:

Layer	Tech	Role
m1/	Python, scikit-learn, LightGBM/XGBoost/CatBoost, SHAP	4 binary classifiers + tests
api/	FastAPI (connected; port 8001)	POST /api/predict
web/	React 19, TypeScript, Vite, React Three Fiber 9, Zustand, Tailwind 4	form, 3D viewer, dashboard

Data flow: *patient form* → *Zustand store* → *typed API client* → (*model response*) → *dashboard* + *3D colouring*. The client also has a documented mock mode that returns a canned worked example through the identical types, so the UI can run offline for demos — the shipped flag talks to the live service.

2. Dataset & preprocessing

Source	UCI ML Repository id 411 , <i>Extension of Z-Alizadeh Sani</i>
DOI / licence	10.24432/C5461K · CC BY 4.0 (unmodified file ships in repo)
Size	303 patients × 59 columns, 0 missing values
Targets	Cath (CAD), LAD, LCX, RCA — binary labels

m1/src/data.py is the single entry point:

- Typed columns.** Numeric vs categorical is decided by explicit lists, not dtype guessing (Region RWMA is forced categorical — it is a region *label*, never a coordinate).
- Constant columns dropped.** Exertional CP is identical for all 303 patients → zero information. 59 – 4 targets – 1 constant = **54 model features** — the same 54 shown in the UI form.
- Targets mapped** to 1/0 with unknown-label guards (any unexpected string fails loudly).
- Target-leakage guard.** Cath, LAD, LCX, RCA are *excluded from every model input* — they are the answers (what the catheter found), not features. Enforced by m1/tests/test_leakage.py; UCI's own dataset note requires the same.

Preprocessing pipeline (inside each model's Pipeline, so train/serve cannot diverge): numeric → median impute → standard scale (logistic baseline); categorical → most-frequent impute → one-hot with handle_unknown="ignore".

3. Model architecture & training

Four **independent binary classifiers**, one shared preprocessing pattern (decision D6):

- Candidates:** LightGBM, XGBoost, CatBoost (class_weight / auto_class_weights="balanced" — n = 303, class imbalance handled by weights, not resampling) + a **logistic-regression baseline** reported for context.
- Selection:** best mean **ROC-AUC** on **5-fold stratified CV**, seed 42.
- No held-out test set** — 303 rows is too few to waste; every metric is either mean ± std over the 5 folds or pooled out-of-fold (each patient predicted exactly once).

- **Operating point:** each outer training split picks its threshold on an inner 3-fold CV (maximising F1) so the validation fold never tunes its own cut-off. Production threshold = max-F1 on pooled OOF (e.g. Cath 0.27, LAD 0.35, LCX 0.25, RCA 0.08). Tuning **raises recall for screening** — recorded separately, never mixed with default-0.5 numbers.
- **Explainability:** SHAP `TreeExplainer` per selected tree model (requirement 3b allows SHAP *or* LIME; decision D5: SHAP only). Model selection is restricted to explainers that report in **probability units** — models that fall back to raw/log-odds contributions are excluded, so all four dashboards are directly comparable.
- **Artifacts:** joblib bundle per target + `metrics.json` + SHAP report, rewritten by `python ml/src/train.py` in ~70 s, versioned with git commit.

4. Evaluation results

5-fold stratified CV, seed 42, pooled out-of-fold, 0.5 threshold unless noted:

Target (model)	ROC-AUC (CV mean ± sd)	OOF accuracy	OOF precision	OOF recall	OOF F1	Recall @ tuned cut-off
Cath / CAD (CatBoost)	0.913 ± 0.052	0.855	0.887	0.912	0.900	0.949
LAD (CatBoost)	0.847 ± 0.047	0.782	0.797	0.842	0.819	0.904
LCX (CatBoost)	0.736 ± 0.076	0.710	0.640	0.597	0.617	0.815
RCA (CatBoost)	0.704 ± 0.050	0.660	0.558	0.465	0.507	0.860

Context: the logistic-regression baseline reaches ROC-AUC 0.925 on Cath — slightly above CatBoost's 0.913 within noise — but tree models were kept because the interpretability requirement is served by `TreeExplainer`. **LCX and RCA are the genuinely hard targets** (fewer positives, subtler signal); we report them at face value rather than hiding them behind threshold tricks. Full per-fold numbers: `ml/artifacts/metrics.json`.

5. Interactive 3D pipeline

Requirement 2a allows Three.js / R3F / VTK.js and *permits* open-source meshes without requiring them; decision D1 kept the anatomy **fully procedural** — zero downloaded assets, so the viewer works offline and stays under the no-GPU budget.

- **Geometry** (`web/src/vessels.ts`, `Heart.tsx`). The heart is one lathe profile (96 segments, two-octave surface grain) with a shared tilt applied to every child so muscle, coronaries and labels can never drift apart. LAD/LCX/RCA are tube geometries swept along anatomically placed curves that ride the sulcus grooves; nine clickable node markers satisfy the literal "coronary artery nodes" wording (decision D3); auricles, aorta, pulmonary trunk, venae cavae, the great cardiac vein and coronary sinus complete the silhouette.
- **Materials** (`materials.ts`). Shared physical materials with a fresnel rim injected via `onBeforeCompile`, procedural roughness + bump tissue map, clearcoat + emissive glow on the vessels; an IBL environment built from local lightformers (nothing fetched), warm back-rim key light, a deep midnight radial backdrop, bloom + vignette in a lightweight post-process chain.
- **Motion**. A systole contraction (radial bulge $\approx 2\times$ long-axis, torsional sway) runs in one `useFrame`; toggleable from the UI. Travelling light bands scroll base-to-apex along LAD/LCX/RCA and the venous return — an emissive pulse injected into the physical shader, so blood visibly flows without particles.
- **Risk mapping**. Each vessel's colour ramp and node colour are driven by the predicted probability shared through the store — green-amber-red is reserved for risk only. Labels (e.g. "LAD 94%") render as DOM pills anchored to 3D positions via a portal.
- **Interaction**. Orbit, zoom, click tube/node to select (second click clears), four camera presets (Anterior/Posterior/Basal/Apical), and one detail panel scoped to the selection — what the vessel supplies plus its SHAP drivers next to the patient's measurements, shown once.
- **Performance**. DPR capped at 1.75, low-poly geometry, single render loop, no external textures — responsive in a modern browser without a dedicated GPU.

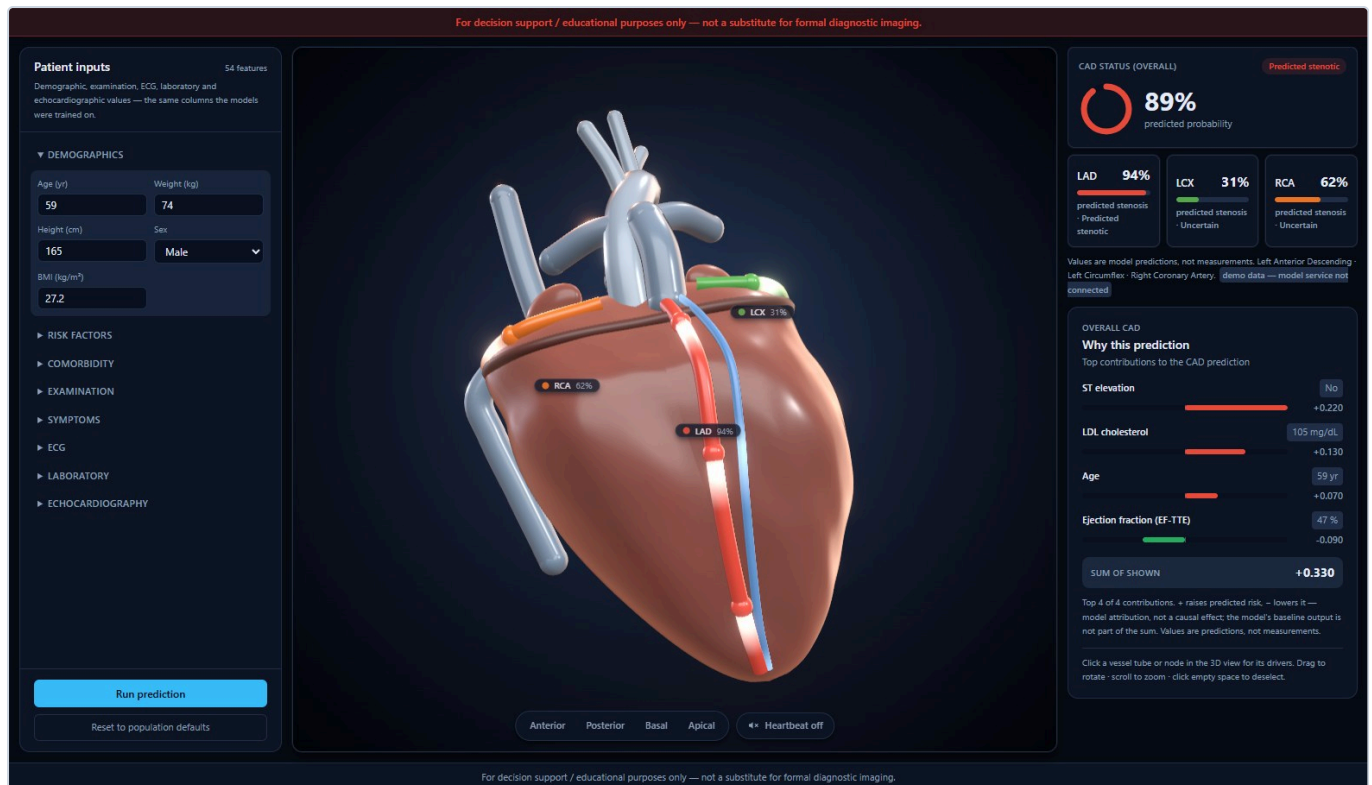


Figure 1 — CardioScope in use: the 54-feature patient form (left), the interactive 3D heart with vessel risk labels, selection and camera presets (centre), and the prediction dashboard with CAD status, per-vessel predicted stenosis probability and SHAP explanation (right).

6. Clinical dashboard & explanation

Beside the canvas (requirement 3): a CAD status card with predicted probability as a donut gauge, three vessel cards (LAD/LCX/RCA with predicted stenosis probability and band), and a single explanation panel — "why this prediction" — where each top contribution appears once, pairing the feature with the patient's measurement, its signed contribution and a direction bar. Selecting a vessel in 3D reframes that same panel to the vessel (what it supplies plus its drivers); the probabilities stay in the cards, never repeated. Missing inputs display "—" rather than invented values. Disclaimers appear as a startup modal, a persistent banner and a page footer (requirement 5).

7. Getting started

```
git clone https://github.com/vaibhavitej-a11y/CardioScope.git
cd CardioScope
python -m venv .venv
.venv\Scripts\activate
pip install -r requirements.txt
cd web && npm install && cd ..

python ml/src/train.py          # trains 4 models, writes ml/artifacts/
npm --prefix web run dev        # frontend → http://localhost:5173
cd api && uvicorn app.main:app --reload --port 8001 # → :8001/docs
```

Prerequisites: Python ≥ 3.11 , Node ≥ 20.19 , Git. Verification:

```
npm --prefix web run lint      # oxlint — clean
npm --prefix web run build     # tsc -b && vite build — passing
python -m pytest               # ml + api — 38/38
```

8. Safety, honesty & reproducibility

- Disclaimer text is enforced in one place (`api/app/constants.py`) and rendered in modal + banner + footer.

- We only claim **predicted probabilities**; the dataset contains no lesion coordinates, so the UI never presents pixel-level lesion mapping — vessel granularity is the resolution the data supports (decision D4).
- Leakage is blocked by tests, not conventions; seeds are fixed (42); model selection, thresholds and variance are all recorded in machine-readable artifacts; the raw dataset file is committed unmodified with CC BY 4.0 attribution.

9. Limitations & future work

- **LCX / RCA discrimination is modest** (ROC-AUC $\approx 0.74 / 0.70$). These are few-positive, subtle-signal targets in a 303-row dataset; we report them at face value. More data or richer inputs would help more than tuning.
- **The prediction service is local-only** (FastAPI, `api/`); it is wired to the UI in development via the Vite proxy, and the typed mock client remains as a one-flag offline fallback (decision D8).
- **Local-only execution** (decision D8) — no hosted deployment; the Devpost requirement is a public repo, not a live URL.
- **The anatomy is schematic**, not patient-specific morphology, and the dataset carries no lesion coordinates — so predictions map to *vessels*, not to positions along a vessel (decision D4).
- **Future work**: calibration curves, prospective validation, imaging modalities as inputs, and torso context (decision D2 — stretch).

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